Mate-in-three: verification and search results

Prepared for Akbar Aman with OpenAI Codex assistance
For Professors Evangelos Kobotis and Drew Shulman; meeting October 14, 2026, 8:30 PM America/Chicago.

Finding

All 20 corrected diagrams have exactly one first move that forces mate within five plies, and none permits a forced mate within one or three plies. Three-ply Minimax selected that key in **6/20** cases; MCTS selected it in **9/60** runs (three seeds per case). Five-ply baseline Minimax completed only **2/20** searches under a 20-second ceiling. Its 18 timeouts are censored searches, not completed five-ply results.

Source, conventions, and independent verification

Only the user-supplied corrected `Positions2.pdf` was used; the earlier erroneous set was excluded. Manual transcription from the four rendered pages was checked against all 1,280 PDF glyph squares. Piece colors, coordinates, side markers and captions agree. All 20 FENs pass `python-chess` validity checks; only case 3 is Black to move. These are structural legality checks, not proofs of historical reachability or verification of the named games.

A mate in three attacking moves can require A, D, A, D, A: **five plies**. The unchanged baseline's `depth=3` counts three individual plies. A separate exact AND/OR solver, independent of both baseline search implementations, tests every legal root move at horizons 1, 3 and 5. It uses

$$W(s,d) = \begin{cases} 1, & \text{defender is checkmated,} \\ 0, & \text{other terminal state or exhausted horizon,} \\ \bigvee_{m \in L(s)} W(s_m, d-1), & \text{attacker to move,} \\ \bigwedge_{m \in L(s)} W(s_m, d-1), & \text{defender to move.} \end{cases}$$

Memoization and logical short-circuiting preserve exact answers; quiet moves and every promotion are included. Separately checked strategy certificates cover *every* defensive reply for each winning key. The oracle and certificate checker share the baseline's rules library, so they are independent search logic, not an independent rules engine.

FENs assume no castling or en-passant rights, halfmove clock zero, and no prior repetition history. All six alternative Black castling-right assignments for cases 1 and 4 preserve the unique key. No other diagram supports castling or a legal immediate en-passant capture. Optional draw claims are not modeled. Historical rights and clocks are not supplied by the diagrams.

Measured protocol and aggregate results

Unchanged September baseline commit `046a8d3`: plain Minimax with material leaves (P=1, N=B=3, R=5, Q=9; mate=±10000); UCT MCTS with $C = \sqrt{2}$, uniform random rollouts capped at 40 plies, material-based cutoff rewards, and most-visited root selection. Each MCTS run requests 1,000 simulations with seed 0, 1 or 2; this is *not* fixed-depth search. A rollout cap is separate from the selected tree path. Each search is capped at 20 seconds; each oracle call at 120 seconds. Calls run serially in case order on Windows, Python 3.12.14, chess 1.11.2. Timings include search overhead but exclude imports and output writing.

Method	Completed	Key hits/completed	Timeouts	Median seconds[†]
Minimax, 3 plies	20/20	6/20	0	0.535
Minimax, 5 plies	2/20	2/2	18	8.195
MCTS, 1,000 simulations	60/60	9/60	0	2.724

[†]Completed calls only; the five-ply median describes just cases 16 and 18. No equal-time or general playing-strength comparison is implied.

All 20 positions: expected and selected first moves

Standard algebraic notation; **bold selections** match the verified key. An asterisk marks a timeout's partial/fallback move and does not indicate a completed search. Full FENs, individual runtimes, scores, node counts and rollout statistics are supplied in the CSV/JSON results.

Case	Verified key	Minimax 3	Minimax 5	MCTS 0	MCTS 1	MCTS 2
1	Nxg7+	Nf4	Bf8*	Bb4	**Nxg7+**	g5
2	Bf5+	Be2+	Bxh7+*	fxe7	fxe7	Bf1+
3	Ne2+	Bc3	Kh7*	Rxg2	Nh1	Bc5
4	Qd8+	Qe3	**Qd8+***	Qxe4	Qxe4	a3
5	Rxg3	Nf3	Re7*	**Rxg3**	hxg3	**Rxg3**
6	Qc8+	**Qc8+**	**Qc8+***	Qxe6	Qxe6	Qxe6
7	Ng5	**Ng5**	**Ng5***	Kg2	Qe3	Qf3
8	Qxg8+	Qf8	Ne7*	Ne7	Qb7	Qf4
9	Rxg8+	Qd6	**Rxg8+***	Qf4	Qa6	Rb2
10	Rh8+	Qxh2+	**Rh8+***	Qc5	Qa6	Qf8
11	Rd8+	Qe8+	Ne7+*	Qxb8	Rxb4	Qe1
12	Rxg7+	Ne7	**Rxg7+***	Qb6	Qg1	Rd7
13	Nf6	Kxd1	**Nf6***	Kxd1	Kxd1	Kxd1
14	Nd6	hxg7+	Qxg7+*	Qe3	Kh2	Kf1
15	Rxa8	**Rxa8**	Be7*	**Rxa8**	**Rxa8**	**Rxa8**
16	Rf7	Kg4	**Rf7**	Rg1	Kf4	Ra1
17	d8=Q+	**d8=Q+**	Rh6*	Ra6	a8=Q+	Rf2
18	Rxg7	**Rxg7**	**Rxg7**	**Rxg7**	Kb1	**Rxg7**
19	Qg8+	**Qg8+**	**Qg8+***	d7	**Qg8+**	Qe2
20	c8=Q+	Rxa7	Re6*	Rxa7	Rxa7	Rxa7

Interpretation and limits

Choosing a key is not proving a mate. Three-ply Minimax cannot see the full five-ply mate here; its six key matches arise from its truncated evaluation and tie-breaking. MCTS's nine matches likewise establish first-move agreement, not a forced continuation. All 60 MCTS calls completed their simulation budgets, but 89.6% of their 60,000 rollouts hit the cap and used a material heuristic. These rewards are not calibrated win probabilities.

Five-ply cost is a material result. Only cases 16 and 18 completed, and both found their verified keys. Of the 18 timeout moves, eight happen to match the key; these are excluded from completed-search success. The 10/20 total returned-key matches must not be described as 10 solved cases. No timing trials were repeated for Minimax; three MCTS seeds are too few for broad statistical claims. Desktop load and fixed execution order can affect timings.

Validation and next step. Sixteen automated tests passed; audit checks covered all 160 raw records (60 oracle calls and 100 baseline searches), source hashes, legal selected moves, and MCTS accounting. Three fixed-seed replays reproduced all root statistics. The strategy certificates prove the winning branches; the oracle separately excludes other keys and faster mates. A useful next experiment is alpha-beta pruning and move ordering on a new branch, followed by a larger MCTS budget sweep. Neural-network training was not run: the supplied `Chess_AI_Python.txt` is retained only as subsequent-scope context.

Provenance and reproducibility

Codex transcribed, implemented the new verifier and experiment scripts, executed the runs, and drafted this report. Akbar's personal review is pending; this does not claim independent student authorship. The September algorithms and artifacts are unchanged. Source files, pinned dependencies, raw logs, proof trees, tests, reproduction commands, meeting notes and an *unsent* email draft accompany this report.

Repository: ak23bar/murl-chess: october14-corrected-positions.

References: Corrected `Positions2.pdf`, user-supplied, four pages; September baseline source and `references.bib` (AIMA Minimax, Kocsis–Szepesvári UCT, Browne et al. MCTS); python-chess core documentation, accessed October 14, 2026.